

AI-Assisted Development Log

Learning without Forgetting Project — Model Optimization

Sami Shamoon College of Engineering

June 2026

Abstract

This document presents selected representative conversations with an AI assistant during the optimization phase of the Learning without Forgetting (LwF) project, for the ImageNet to Scenes experiment. The purpose is not to include every message, but to document the main AI-assisted interactions that led to a concrete improvement over the standard LwF method. The log follows the reasoning chain we actually used: implementing a cosine-normalized classifier from a reference article, reading the resulting numbers, diagnosing why the new task underperformed, and progressively tuning the method until it improved both the new task and the old task at the same time. All training runs, measurements, and final results were executed by the student; the AI assistant served as a technical guide, debugging assistant, and documentation support.

Contents

- 1 Introduction
- 2 Conversation 1: Adding a Cosine-Method Optimization to LwF
- 3 Conversation 2: Selecting a Single Method — the Cosine-Normalized Classifier
- 4 Conversation 3: Running Only the New Block Without Re-Training
- 5 Conversation 4: First Result — Old Task Up, New Task Down
- 6 Conversation 5: Diagnosing the Drop — the Scale Parameter
- 7 Conversation 6: Boosting the New-Head Learning Rate
- 8 Conversation 7: Final Result and Method Selection
- 9 Summary of AI Contribution
- 10 Conclusion
- 11 AI Assistance and Credit

1 Introduction

This optimization log documents the second half of the project, where the goal moved from reproducing Learning without Forgetting to improving it. The experiment used ImageNet classification as the old task and MIT Indoor Scenes (67 classes) as the new task, on an AlexNet backbone. The baseline methods (Fine-Tuning, Feature Extraction, and standard LwF) were already implemented and measured. The objective of this phase was to propose and validate a real, meaningful improvement to the LwF method while keeping the original training protocol unchanged so the comparison would remain fair.

The starting reference points were standard LwF (ImageNet 59.82%, Scenes 56.27%), and the fixed protocol used throughout: a head-only warm-up, then a joint LwF stage with a Knowledge-Distillation loss on the old task, fixed seed 42, a five-epoch budget, and an 80-batch subset for ImageNet evaluation due to compute limits.

2 Conversation 1: Adding a Cosine-Method Optimization to LwF

Student Question

I have my LwF notebook for the ImageNet to Scenes experiment. Can you add an optimization based on the cosine method, as a new block I can run, without changing the existing code or its protocol?

AI Response

The phrase "cosine method" is ambiguous in an LwF context, so the assistant first separated the two reasonable meanings: a cosine-annealing learning-rate schedule (an optimizer change that leaves the loss untouched), and a cosine-similarity distillation loss (a change to how the old task is preserved). Both were implemented as a single self-contained block appended to the notebook, reusing the existing warm-up, seeds, loaders, and training functions so the original protocol stayed intact.

Impact on the Project

This established the working pattern for the whole phase: every optimization would be a separate, self-contained block that does not modify earlier cells, so results already obtained remain valid.

3 Conversation 2: Selecting a Single Method — the Cosine-Normalized Classifier

Student Question

I do not want two optimizations. Use only one method — the one explained in the article I uploaded — and let me clearly see the improvement at the end of the code.

AI Response

The uploaded article described a specific method: the cosine-normalized classifier. A standard classifier scores a class with the dot product, which mixes how well an image matches a class with how large that class template is. New-class templates tend to grow larger, biasing the model toward new classes. The cosine classifier removes the vector norms and decides only by direction, then rescales the result with a single learnable scale so the softmax keeps a usable range:

$$z = \eta \cdot \cos(f, w) = \eta \cdot (f \cdot w) / (\|f\| \cdot \|w\|)$$

Because the old and new tasks use separate heads in this notebook, the assistant applied the cosine head to the new (Scenes) head, where the template-norm growth actually occurs, and left the old ImageNet head as the original linear classifier so the Knowledge-Distillation memory loss stayed exactly as in standard LwF. A final

cell printed the new method next to the baselines so the improvement was visible directly.

Impact on the Project

This fixed the scope of the work to a single, well-motivated method that changes only the classification head, and added an explicit comparison readout for the new versus old task.

4 Conversation 3: Running Only the New Block Without Re-Training

Student Question

Can I run only the new block? I do not want to re-run everything, because that would waste time and I want to keep my previous results.

AI Response

The assistant explained that the new block depends on definitions and data created in the earlier setup cells, so it cannot run in complete isolation, but the expensive training of the three baseline methods happens in a single cell that can be skipped. The block was made robust so it could run after only the setup and definition cells: it creates the results list if it is missing and falls back to the previously recorded baseline numbers when the live values are not in memory.

Impact on the Project

This let the optimization be iterated quickly without re-training the baselines, while keeping the previously obtained numbers intact for comparison.

5 Conversation 4: First Result — Old Task Up, New Task Down

Student Question

I ran the cosine classifier. It improved the old task but the new task got worse. I want to try another optimization that improves on this, and it should still be related to the cosine method.

AI Response

The first cosine result improved the old task (ImageNet 60.74%, an increase of 0.92 points) but reduced the new task (Scenes 52.31%, a drop of almost 4 points versus standard LwF). The assistant treated this as informative rather than a failure: the norm-fairness idea clearly helped preservation, so the question became why the new task suffered, and how to fix it without abandoning the cosine method.

Impact on the Project

This reframed the next step as a targeted diagnosis of the new-task drop instead of switching to an unrelated technique, keeping the work focused.

6 Conversation 5: Diagnosing the Drop — the Scale Parameter

Student Question

Why did the new task get worse, and how can I improve it without changing the whole protocol?

AI Response

The assistant read the training log and found the key clue: the Scenes validation accuracy rose steadily across all five epochs (44, 46, 47, 48, 50) and was still increasing when the epoch budget ran out. The cosine head had not converged. The cause was the scale: a cosine lives between -1 and 1, so with a small scale the logits stay in a narrow range, the softmax is nearly flat, the gradient is weak, and the head learns slowly and under-confidently. The article's own section on the scale names this exact problem.

The proposed fix was to start the learnable scale higher (30 instead of 16), which sharpens the logits from the first step so the same five epochs reach a confident solution. This changed only one number; the old head, the distillation loss, the seeds and the budget were unchanged. The result was ImageNet 60.51% and Scenes 56.19% — back to parity with standard LwF on the new task while keeping the old-task gain, a genuine improvement on both axes relative to the previous cosine run.

Impact on the Project

This identified the scale as the real bottleneck and produced the first version that did not sacrifice the new task, but the training curve suggested the head could still learn more.

7 Conversation 6: Boosting the New-Head Learning Rate

Student Question

I got a small improvement. How can I improve the results further without changing the whole code? Send me a fixed block I can put in place of the previous one.

AI Response

The diagnosis stayed the same — the new head was still under-trained within the budget — so the assistant targeted the head directly. The single-rate optimizer was replaced by two parameter groups: the shared backbone and the old head kept the base learning rate, while the new cosine head (its weights and its scale) trained at five times that rate. This mirrors the way the original notebook already gives Feature Extraction a higher rate. Because the shared layers still move at the base rate, the drift away from the old task is identical to standard LwF, so old-task preservation is not traded away for new-task accuracy.

The change was delivered as one drop-in cell to paste in place of the previous block, so it could be run in the existing session without losing any in-memory state.

Impact on the Project

This isolated the new head as the component that needed more learning capacity and addressed it with a contained, well-precedented change rather than a protocol change such as more epochs.

8 Conversation 7: Final Result and Method Selection

Student Question

This improved the results. Explain the final method, why I chose it, the path we took to optimize it, and how the results could be improved further in the future.

AI Response

The final method is a cosine-normalized classifier on the new task with a raised learnable scale and a dedicated higher learning rate for that head only, layered on top of an otherwise unchanged LwF pipeline. Its measured result was ImageNet 60.72% and Scenes 58.06%.

Compared with standard LwF, this improves both tasks at the same time: the new task by 1.79 points (56.27 to 58.06) and the old task by 0.90 points (59.82 to 60.72). Compared with the plain cosine classifier, it recovers and then exceeds the new task by 5.75 points with essentially no change to the old task. It does this without touching the LwF protocol — same epochs, same loss, same warm-up, same seeds — changing only the head's geometry, its scale, and its learning rate. Improving both axes at once is exactly the hard case the original paper emphasizes, which is why this version was selected as the project's proposed improvement.

For future work, the assistant suggested directions grounded in the same diagnosis — the new head wants more learning capacity, the backbone must be protected: a small grid over the scale and the head learning-rate multiplier; allowing the head to fully converge by relaxing the epoch budget (applied equally to the LwF baseline for fairness); trading along the old/new curve by lowering the distillation weight; stronger augmentation on the new-task images only; adding a cosine margin once the head is well-trained; and averaging over several seeds to separate real gains from noise before further tuning.

Impact on the Project

This produced the final reported method and its justification, and framed the remaining headroom as concrete, fairness-preserving next steps.

9 Summary of AI Contribution

During the optimization phase, the AI assistant was used for the following purposes:

- clarifying the meaning of a "cosine method" and proposing concrete variants;
- implementing the cosine-normalized classifier from the reference article as a self-contained block;
- making each block runnable without re-training the baselines, while preserving earlier results;
- diagnosing, from the training logs, why the new task underperformed (the scale parameter);
- proposing the higher scale and the boosted new-head learning rate as targeted, protocol-preserving fixes;
- designing the comparison readout that reports old-task and new-task changes against each prior method;
- preparing the final explanation, the future-work directions, and this documentation.

All experiments, training runs, measurements, and final results were executed by the student. The AI assistant was used as a technical guide, explanation tool, debugging assistant, and documentation support.

10 Conclusion

The selected conversations show how the optimization phase progressed by reading each result and fixing the specific weakness it exposed. Beginning from the cosine-normalized classifier described in the reference article, the work moved through a clear diagnostic chain — identifying the scale as the cause of slow, under-confident learning, then giving the new head enough learning capacity to converge within the fixed budget. The final method improves both the new task and the old task relative to standard LwF without altering the training protocol, which makes it a fair and meaningful improvement. The AI interactions were most useful for interpreting training behavior, proposing minimal and well-motivated changes, and keeping every step reproducible and comparable.

11 AI Assistance and Credit

In accordance with academic integrity and responsible-use guidelines, the role of artificial intelligence in this work is disclosed in full. The optimization phase documented above was carried out with the assistance of Claude, a large language model developed by Anthropic. The assistant contributed to clarifying the cosine-classifier method, implementing it as self-contained code blocks, diagnosing training behavior from

logs, proposing the scale and learning-rate refinements, and preparing this documentation.

The intellectual decisions, the choice of which suggestions to adopt, and the verification of every claim against measured results remained with the student. All datasets were prepared by the student, and all training runs, measurements, and final reported values were executed and validated by the student on their own computing environment. The AI assistant functioned as a technical guide, debugging aid, and writing support, and not as a source of experimental results.

AI Tool Credit

Tool: Claude (Anthropic) · Role: technical guidance, code implementation, debugging, and documentation support · Oversight: all outputs reviewed, executed, and verified by the student.